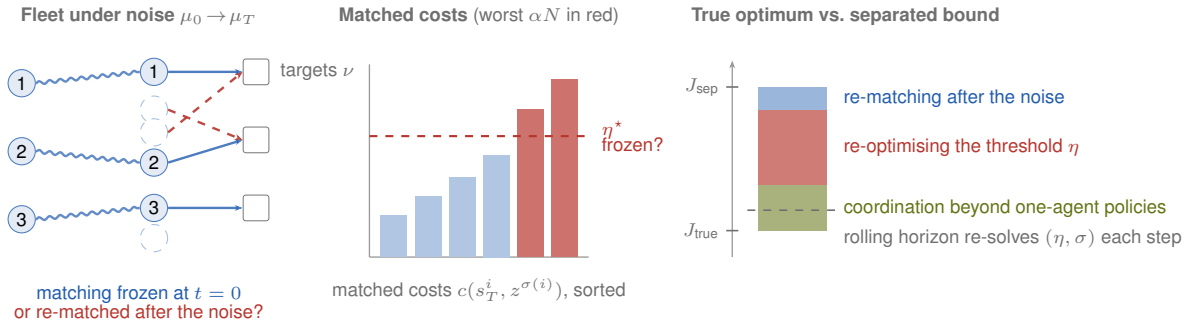


Project type: SA (semester project)
Project title: Risk-Sensitive Separation for Fleet Control

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Supervising professor: Autumn 2026

Semester:
Start date - end date:
Office space:



Left: a fleet moves under noise; the separated controller fixes the matching (and the threshold) before the noise is realised, the true optimum may re-decide afterwards. Middle: the fleet is scored on the worst α -fraction of the matched costs, whose Rockafellar–Uryasev threshold η^* is the one extra shared variable. Right: the project computes both values exactly on small instances and splits their gap into three interpretable parts.

Project description. Emergency response, like ambulance and fire services are not only scored on their *average* response time but also on its *tail*. For example, NHS England scores ambulance trusts per call category on a mean and a 90th centile. For Category-2 calls the standards are 18 and 40 minutes, while the measured performance in May 2026 was a mean 29 min 13 s with a 90th centile of 59 minutes, 19 min over the standard [4]. This combination makes the problem hard. The two objectives can disagree and only improving the average can leave the tail of the distribution unchanged or even worsen it. On top of this, studying the tail of the distribution of outcomes across the fleet is a nonlinear functional of the whole population. Which calls have the largest response time depends on the position of all units and on which unit serves each call. Solving this control problem directly requires dynamic programming over all possible fleet configurations, which quickly becomes too expensive as the fleet grows.

In recent work in our group we proved a *separation principle* for fleet control and turned it into the SALT family of algorithms (Separation-based Assignment and Learning via optimal Transport) [5, 10]. For an *average* objective, the population cost-to-go is bounded above by an optimal transport problem whose transport cost is the *single-agent* cost-to-go $\hat{J}_t(s, z)$ of reaching target z from state s . One learns a single target-conditioned policy, cheap and independent of the fleet size, and coordinates the fleet

by solving an assignment problem using the single-agent expected costs as transport costs. The bound is exact under deterministic dynamics and its only slack under noise is that the assignment is blind to future noise.

This project studies whether the same separation principle still holds when we care about improving the largest costs across the fleet. We focus on the conditional value-at-risk (CVaR), which measures the average of the worst α -fraction of costs (e.g. $\alpha = 0.1$ corresponds to the average of the worst 10%). This relates to the 90th percentile discussed above, but also accounts for how large the costs beyond that percentile are. Additionally, the following Rockafellar-Uryasev representation [8] of CVaR seems to suggest the possibility of this extension.

For matched costs $c_1, \dots, c_N$,

$$\text{CVaR}_\alpha(c_1, \dots, c_N) = \min_{\eta} \left\{ \eta + \frac{1}{\alpha N} \sum_{i=1}^N (c_i - \eta)_+ \right\},$$

where $(x)_+ = \max\{x, 0\}$. Once the threshold η is fixed, the sum contains a separate term for each unit. This suggests solving single-agent control problems with these modified costs, using their expected values to assign units to targets, and then minimising over the threshold. A procedure that deeply resembles the one used for the mean objective. We hence conjecture that the resulting separated value J_{sep} is an upper bound on the optimal fleet cost J_{true} , with equality under deterministic dynamics. The first aim of the project is to determine whether this conjecture is correct and which assumptions it requires. If this bound holds, the next question is whether it is close enough to the optimum to be useful, and whether updating the assignment and threshold during operation improves the resulting controller. A first smoke test on a 3×3 grid with two units seems to be positive: the bound held on every one of 200 exact instances, was exact without noise, and its gap grew from 0.005 to 0.42 (on a true value of 1.1) as the action noise grew from 5% to 50%; at the same noise it was about five times larger for the tail objective than for the mean.

Motivation. Developing a control method to improve real-world response times requires coordinating many units under uncertainty in a time-varying scenario. Approximate dynamic programming makes fleet control tractable by simplifying the value function of the whole fleet. Its accuracy depends on which information is retained and how the approximation is fitted [?, ?]. Risk-sensitive MARL methods also learn a team-level approximation from individual agent values. However, combining individual risk measures need not recover the team's risk measure, and even methods designed to ensure consistent decisions retain approximation and learning errors [?, 9]. We, instead, aim to reduce reduce this burden by *deriving* a principled, scalable and interpretable separation principle, combining a single-agent control problem with an assignment transport one.

Approach. You will start from the existing separation principle for average objectives and investigate whether it extends to CVaR. The first step is to define the fleet objective precisely and identify the assumptions on the dynamics and the information available to each unit.

The Rockafellar-Uryasev representation of CVaR suggests a possible construction: fix a common threshold η , solve the corresponding single-agent control problems, and coordinate the units through an assignment. The main task is to determine whether this construction gives an upper bound on the optimal fleet cost, and under which conditions the two values coincide. We will investigate the role of random disturbances, how fixing the threshold and assignment affects the argument, and whether any resulting gap can be explained mathematically. Experiments and benchmarking will support this analysis.

Goals.

1. Determine whether the separation principle extends to the finite-state CVaR objective. State the assumptions precisely and either prove the proposed upper bound or construct a counterexample that explains why it fails. Investigate separately whether the separated value equals the optimum under deterministic dynamics.
2. Build a tested implementation that computes the optimal fleet cost, the separated bound, and the cost achieved by the separated controller. Include comparisons with the mean, a combination of

mean and CVaR, the probability of exceeding a cost threshold, and a chosen percentile.

3. Identify the conditions under which the bound is close to the optimum and those under which the gap is large.
4. If time allows, expand the numerical results to larger fleets with potentially real-world data.

Qualifications. We are looking for a highly motivated student with:

- A strong mathematical background in probability, linear algebra, and analysis. Familiarity with stochastic processes is a plus.
- Some exposure to optimal control, dynamic programming, or reinforcement learning, and an interest in theoretical research.
- Programming skills in Python to implement small examples and explore theoretical ideas.
- The ability to work independently and communicate mathematical arguments and results clearly in writing.

Familiarity with optimal transport or risk measures is welcome, but can be developed during the project.

How to apply. Please send your CV and transcript to lpezzetti@ethz.ch, together with a couple of sentences on which part of the project appeals to you. Happy to discuss informally before you decide.

References

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